Efficient Hyperspectral Multiclass Change Detection Based on Spectral–Spatial Harmonization

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***Abstract*—The rich spectral information in hyperspectral images (HSIs) provides a significant advantage for fine-grained, multiclass change detection (CD), yet a robust and practical framework remains elusive. Existing deep learning (DL) methods are often fragmented, failing to simultaneously address a triad of critical challenges: *1) spectral inconsistency* between HSI pairs acquired from different sensors or times; *2) inefficient static fusion* of hierarchical spectral–spatial features, which struggles to model complex changes; and *3) severe label scarcity* for multiclass supervision, which limits model training. To systematically solve these interconnected problems, we propose the deep differential feature fusion (D^2F^2) framework, a holistic, end-to-end architecture. D^2F^2 introduces two novel components: a spectral adaptive sampling (SAS) module that adaptively remaps inputs with arbitrary spectral bands to a unified feature space, thereby alleviating spectral inconsistency caused by missing spectral channels and band-number mismatch. Concurrently, a spatial hierarchical aggregation (SHA) module replaces static fusion with a dynamic, attention-based mechanism to generate highly discriminative, change-aware features. To address label scarcity, the entire framework is trained using an efficient semi-supervised strategy. This fully convolutional design also enables ultraefficient map-to-map inference, which eliminates the massive computational redundancy of patch-based methods. Extensive experiments on three challenging HSI datasets demonstrate that D^2F^2 not only achieves new state-of-the-art accuracy but is also orders of magnitude more computationally efficient than competing DL models. The code will be available at: https://github.com/dmhz/HSCD.**

***Index Terms*—End-to-end, hyperspectral images (HSIs), multiclass change detection (CD), semi-supervised learning, spatial hierarchical aggregation (SHA), spectral adaptive sampling (SAS).**

I. Introduction

CHANGE detection (CD) is the process of identifying change regions on the Earth's surface by comparing images acquired over the same area at different times [1]. The CD has been developed and successfully used in different remote sensing applications, such as urban landscape transitions [2], [3], land use monitoring [4], [5], and natural disaster assessment [6]. The timely and accurate CD of the Earth's surface features provides a basis for a better understanding of the relationship and interaction between humans and natural phenomena [7].

With the rapid advancement of remote sensing technology, hyperspectral images (HSIs) have become increasingly accessible. Owing to their continuous and narrow spectral bands, HSIs provide much richer material information than conventional multispectral or optical imagery, enabling not only the identification of subtle changes but also the discrimination of different change types [8]. This property makes HSIs particularly attractive for multiclass change detection (CD). However, the high-dimensional nature of HSIs also introduces significant challenges, such as the curse of dimensionality and increased computational overhead [9]. More importantly, although existing methods achieved encouraging results in binary CD, their transition to practical and fine-grained multiclass HSI-CD remains limited by three fundamental challenges that have not yet been holistically addressed.

1) *The Dilemma of Spectral Inconsistency:* In real-world scenarios, HSIs are often acquired by different sensors or under varying conditions, leading to inconsistent available bands between image pairs [10]. Most deep learning (DL) models require fixed-input dimensions, rendering them ineffective when faced with such spectral misalignment [11]. This limitation severely restricts the flexible deployment of CD algorithms across heterogeneous HSI sources. Consequently, current solutions, such as manual band selection or interpolation, are often suboptimal and may introduce bias, highlighting the need for more adaptive input handling mechanisms.
2) *The Inefficiency of Static Feature Fusion:* Convolutional neural networks (CNNs) extract hierarchical features, yet effectively fusing these features for multiclass CD remains nontrivial. Simple concatenation or summation of multilevel features—common in many frameworks [12], [13]—is often suboptimal for capturing the complex spectral–spatial characteristics required for distinguishing various change types [14]. This can lead to compromised performance when

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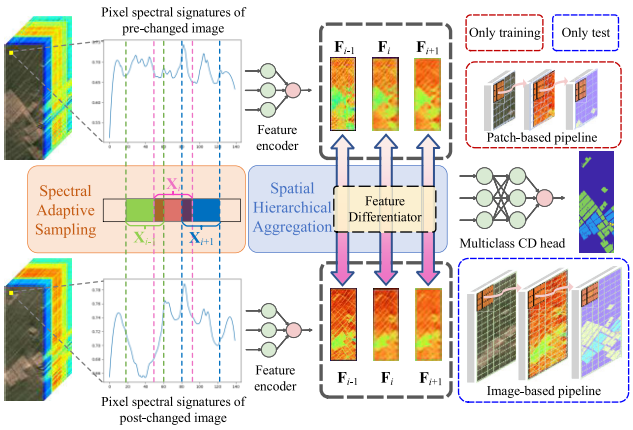


Fig. 1. Architecture of the proposed D^{2}F^2 framework for multiclass CD. It features: 1) an SAS module to handle inconsistent spectral inputs and 2) an SHA module for dynamic spectral–spatial feature fusion. The framework operates in a fully convolutional (map-to-map) manner, enabling efficient inference while maintaining high accuracy.

change characteristics vary significantly, calling for more dynamic and discriminative fusion strategies [15].

3) *The Data Scarcity in a Multiclass Context:* The performance of DL models is heavily dependent on large-scale labeled data. Multiclass CD demands an even greater number of high-quality labels to capture finer distinctions between change categories. Manual annotation of HSIs is expensive and time-consuming, creating a bottleneck for model training [16]. While semi-supervised learning offers a promising direction [17], its effective application in HSI-CD, particularly for complex multiclass tasks where labeled samples are extremely limited, is still underexplored [18].

A survey of existing approaches showed that these challenges are often addressed in a fragmented manner. Predominant DL models [2], [3], [8], [19], [20], [21], including spatial–spectral attention networks like SSA-SiamNet [22] and advanced Transformer- or Mamba-based frameworks [23], [24], achieved impressive results for binary CD, yet their direct extension to multiclass scenarios was limited, as they lacked the specific architectural design needed to discriminate between fine-grained change types. Several pioneering works ventured into multiclass HSI-CD [25], [26], [27], [28], introducing valuable innovations. Therefore, there remains a clear need for a unified and end-to-end trainable framework that can simultaneously handle arbitrary spectral inputs, enable dynamic and discriminative feature fusion, and learn effectively from limited annotations. Addressing these issues in a coordinated manner is essential for improving both the accuracy and the practical deployment of multiclass HSI-CD.

To this end, we present the deep differential feature fusion (D^{2}F^2) framework, a holistic solution that systematically closes this gap. D^{2}F^2 operates through a tightly integrated, end-to-end trainable architecture. Fig. 1 illustrates the overall architecture of the proposed framework. First, rather than relying on simple preprocessing like band selection or image differencing, a spectral adaptive sampling (SAS) module acts as a flexible front end. It jointly processes HSI pairs with arbitrary and differing band numbers, remapping them into a unified, task-aligned, difference-aware feature space, thus decoupling the model from specific sensor configurations. Second, the resulting multilevel differential features are dynamically fused by a spatial hierarchical aggregation (SHA) module. Rather than statically combining features, SHA employs a self-attentive mechanism to selectively aggregate the most discriminative spatial and semantic information from different network depths, tailoring the feature representation for multiclass distinction. Finally, this powerful feature extraction backbone is trained using a novel semi-supervised strategy that strategically leverages limited manual labels and abundant unlabeled data. This strategy creates a virtuous cycle where well-aligned features from SAS facilitate accurate pseudo-label generation, which in turn guides the SHA module to refine its feature selection, enabling the model to achieve robust performance even with sparse annotations. During inference, the fully convolutional nature of D^{2}F^2 enables efficient map-to-map prediction, offering superior computational efficiency over patch-based counterparts.

The main contributions of this work are threefold.

1) We propose an SAS module that enables end-to-end multiclass CD for HSI pairs with arbitrary and inconsistent spectral bands. Different from prior spectral–spatial methods that assume fixed spectral dimensionality or require handcrafted preprocessing, SAS learns a unified, task-aligned, and difference-aware representation, improving spectral comparability and sensor adaptability.
2) We design an SHA module for dynamic multilevel differential feature fusion. Unlike conventional static concatenation or summation, SHA selectively aggregates the most discriminative spatial-semantic cues from different depths, which is critical for fine-grained multiclass change identification.
3) We integrate SAS and SHA into a unified D^{2}F^2 framework with a codesigned semi-supervised learning strategy. Therefore, D^{2}F^2 is not merely a minor extension of existing spectral–spatial techniques, but an end-to-end framework that jointly addresses heterogeneous spectral inputs, dynamic differential fusion, and label-efficient multiclass learning.

II. Related Works

The evolution of hyperspectral CD mirrors broader trends in computer vision, progressively advancing from pixel-level statistical analysis to subpixel-level decomposition and, most recently, to deep, data-driven feature learning. This section critically reviews these paradigms, highlighting their respective strengths and, crucially, the unresolved challenges in multiclass CD that motivate our work.

A. CVA and Its Evolution

Early and foundational CD research is dominated by change vector analysis (CVA) and its variants. The core principle of CVA [29] is to compute a difference vector in spectral

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space, with its magnitude indicating change intensity. Seminal works like multivariate alteration detection (MAD) [30] and its iterative version IR-MAD [31] enhanced robustness to nonlinear radiometric differences. Subsequent innovations focused on refining this framework, for instance, through compressed 2-D representations [32], improved thresholding [33], and increased robustness to acquisition geometry [34].

While CVA-based methods are intuitive and computationally efficient, their performance is inherently tied to the quality of handcrafted features and thresholding rules. They struggle with the high dimensionality and redundancy of HSIs, lack the ability to model complex spatial context, and offer limited discriminative power for fine-grained multiclass change identification [27]. This renders them insufficient for the demands of modern multiclass CD tasks.

B. SU Approaches

Beyond pixel-level differencing methods, hyperspectral CD has also developed a subpixel-level line of research to address the mixed-pixel problem caused by the limited spatial resolution of HSIs. Spectral unmixing (SU) models each pixel spectrum as a combination of endmember signatures and their corresponding abundances. Along this line, Erturk and Plaza [28], [35], [36] pioneered sparse unmixing for CD, while Liu et al. [37] and Jafarzadeh and Hasanlou [38] further explored unsupervised frameworks based on abundance variation analysis. These methods provide physically interpretable descriptions of change at the subpixel level. However, they are often computationally demanding and highly sensitive to endmember extraction and unmixing quality. Moreover, because they are mainly designed for abundance decomposition rather than end-to-end representation learning, they are less effective at modeling complex spatial context and are not fully aligned with the computer vision-driven multiclass CD frameworks that dominate recent research.

C. DL Paradigms in HSI CD

DL has fundamentally reshaped CD by enabling models to learn hierarchical feature representations directly from data. Early studies demonstrated this potential through Siamese architectures for feature comparison [39], [40], recurrent structures for temporal modeling [41], and coupling networks for consistent representation learning [12]. More broadly, this progress in CD reflects the rapid advancement of AI across the hyperspectral imaging domain, where recent models, such as SpectralGPT [42] and LRR-Net [43], have substantially enhanced spectral–spatial feature learning. However, despite these advances, state-of-the-art (SOTA) research in multiclass CD remains fragmented. Existing methods often evolve along separate lines, each addressing only part of the problem rather than providing a unified solution.

One major research thrust focuses on improving feature representation by overcoming the inefficiency of static feature fusion. The field has recently embraced advanced architectures like Transformers and Mamba (state space models) to achieve this. Models such as SST-Former [44] and GTransCD [45] leverage attention mechanisms for global spectral–spatial dependency modeling, while emerging works like SFMS [23] explore Mamba for efficient long-range feature extraction. While these methods represent the forefront of dynamic feature fusion, their representational power comes at a cost: they are notoriously data-hungry, which severely exacerbates the problem of data scarcity in a multiclass context. Critically, their architectures, which rely on fixed-length inputs, still presuppose perfectly aligned image pairs, thus failing to address the dilemma of spectral inconsistency between sensors.

A second, parallel thrust directly tackles this data scarcity bottleneck. This line of work includes unsupervised methods like BCG-Net [26] (which guides multiclass detection using binary change results), weakly supervised networks like WSCDNet [46], and various few-shot learning (FSL) frameworks [47]. While these approaches make significant strides in reducing reliance on annotations, they often do so by employing simpler backbones or relying on multistage pipelines and pseudo-labeling, which can be sensitive to noise. Their focus on data-efficiency often means they compromise on the advanced, dynamic fusion mechanisms needed for highly discriminative multiclass separation. Furthermore, they also operate under the assumption of spectrally consistent inputs.

A third, emerging thrust, often termed heterogeneous CD, explicitly targets the dilemma of spectral inconsistency [48], [49]. Methods like CHMFFN [50] focus on learning domain-invariant feature spaces to handle images from different sensors. However, this specialization on domain alignment often comes at the expense of the other two challenges. These models are not inherently designed to perform highly discriminative hierarchical fusion for subtle multiclass distinctions, nor are they optimized to operate in the data-scarce scenarios that are typical of real-world multiclass CD tasks.

D. Synthesis and Unresolved Research Gap

Our analysis of the literature reveals a highly fragmented landscape where the field still lacks a unified framework. Instead, current SOTA methods tend to offer partial solutions, often specializing into distinct "silos": advanced fusion architectures (e.g., Transformers) excel at feature representation but remain data-hungry and sensor-dependent; data-efficient methods (e.g., FSL and unsupervised) reduce label dependency but compromise on fusion power and ignore spectral inconsistency; and methods designed for spectral mismatch or missing-channel conditions improve robustness to varying band availability, but are not optimized for data scarcity or advanced multiclass fusion. Therefore, a significant and clear gap remains for a single, end-to-end framework for multiclass HSI-CD that can simultaneously and cohesively address the complete triad of practical challenges: 1) handling arbitrary spectral band inconsistency between image pairs; 2) performing dynamic and discriminative fusion of hierarchical deep features; and 3) achieving robust performance with limited labeled samples in a multiclass context. This gap motivates our proposed D^2F^2 framework. Different from existing hyperspectral CD methods, our framework is not built upon a fixed-band assumption, does not rely on static multilevel fusion, and avoids multistage auxiliary pipelines. Instead, D^2F^2 unifies

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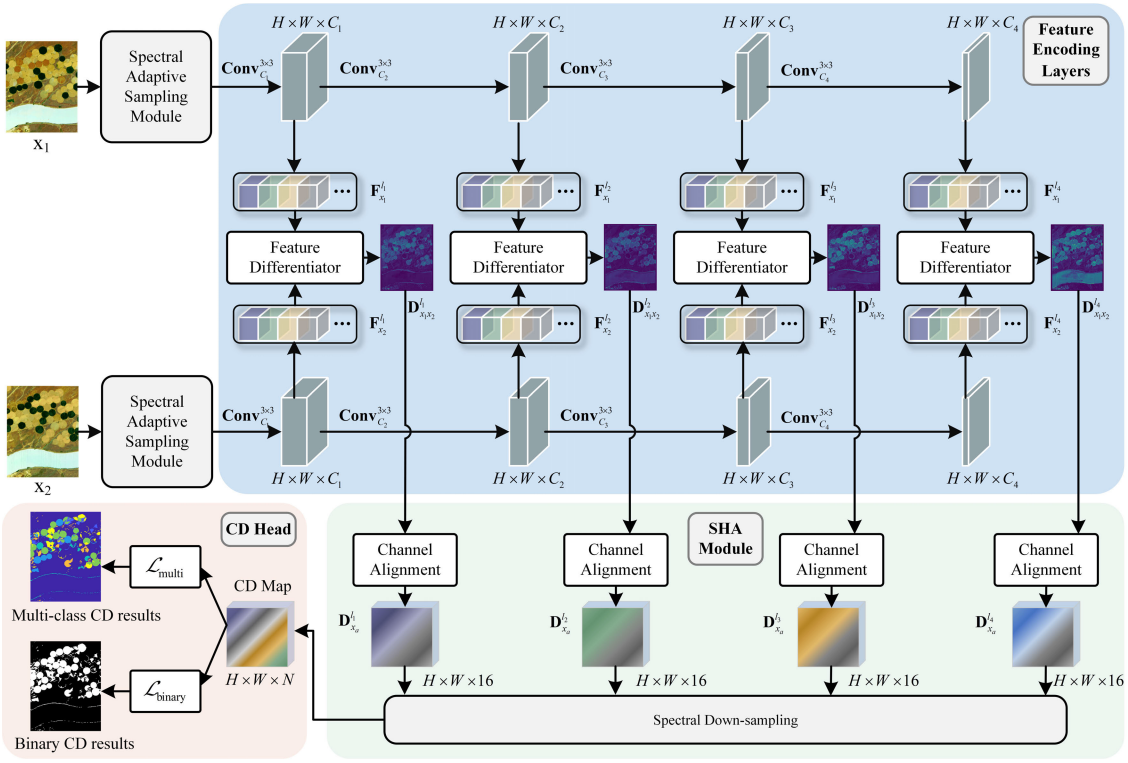


Fig. 2. Architecture of the proposed D^{2}F^2 framework. HSI pairs $(\mathbf{X}_1, \mathbf{X}_2)$ are first projected into a unified feature space by the SAS module. A weight-sharing encoder then extracts four hierarchical feature levels with channel dimensions $C_1 = 128$, $C_2 = 96$, $C_3 = 64$, and $C_4 = 32$. Differential features $(\mathbf{D}_l)$ are computed at each level and then dynamically fused by the SHA module to produce the final multiclass change map via a CD Head.

adaptive spectral alignment, dynamic hierarchical differential aggregation, and semi-supervised optimization within a single end-to-end trainable architecture for multiclass HSI-CD.

III. Methodology

This section details the proposed D^{2}F^2 framework for multiclass hyperspectral CD. We first formulate the problem and analyze the fundamental limitations of traditional differencing approaches in a multiclass context. We then present the overall architecture of D^{2}F^2 and elaborate on the design and implementation of its core components, which are engineered to cohesively solve the primary challenges of spectral inconsistency, inefficient feature fusion, and label scarcity.

A. Problem Formulation

Let $\mathbf{X}_1 \in \mathbb{R}^{H\times W\times C_1}$ and $\mathbf{X}_2 \in \mathbb{R}^{H\times W\times C_2}$ be two coregistered HSIs acquired over the same geographical area at times t_1 and t_2, respectively. H and W are the spatial dimensions, and C_1 and C_2 are the number of spectral bands, which may not be equal. The goal of multiclass CD is to generate a change map $\mathbf{Y} \in \mathbb{R}^{H\times W}$, where each pixel is assigned a label from the set $\Omega = \{\omega_n, \omega_{c_1}, \omega_{c_2}, \ldots, \omega_{c_K}\}$, representing "unchanged" and K distinct "changed" classes, respectively.

B. Overview

The proposed D^{2}F^2 framework, illustrated in Fig. 2, is an end-to-end trainable network designed to systematically address the core challenges of multiclass HSI-CD. Its workflow begins by addressing the critical practical challenge of *spectral inconsistency*. In real-world scenarios, HSI pairs often have differing band counts ($C_1 \neq C_2$) due to cross-sensor acquisition or data corruption, rendering standard fixed-input CNNs unusable. To overcome this, we first introduce an SAS module. This adaptive front end intelligently projects both $\mathbf{X}_1$ and $\mathbf{X}_2$ into a unified, fixed-dimensional feature space $\widehat{C}$.

These aligned feature maps are then fed into a weight-sharing convolutional encoder. The shared weights are critical, ensuring that identical land covers from both time points are mapped to similar feature spaces, which makes the subsequent differencing step meaningful. The encoder extracts hierarchical feature pairs $\{\mathbf{F}_l^1, \mathbf{F}_l^2\}_{l=1}^L$ at L different scales. At each scale l, differential features $\mathbf{D}_l = \mathbf{F}_l^2 - \mathbf{F}_l^1$ are computed. The next challenge is to fuse these multiscale features. Simple static fusion, like concatenation or summation, is suboptimal as it treats all feature levels and spatial locations equally. We address this by designing the SHA module, which performs a dynamic, attention-based aggregation to produce a single, highly discriminative fused feature map. Finally, a lightweight CD Head maps these fused features to the final pixel-wise predictions. To train this deep architecture given the severe *scarcity of multiclass labels*, the entire model is optimized using a novel semi-supervised strategy that synergistically leverages sparse manual labels and abundant unlabeled data.

C. SAS Module

The SAS module (Fig. 3) achieves its input flexibility by adaptively sampling the spectral dimension using s independent 1×1 convolutional groups. Given an input HSI $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$ with an arbitrary number of bands C, we define

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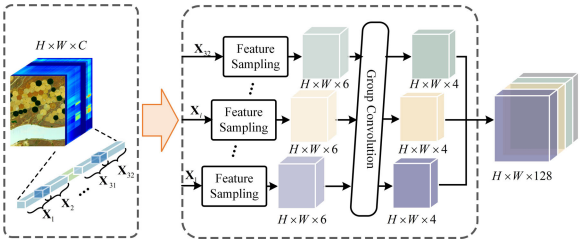


Fig. 3. SAS module. It uses a grouped 1×1 convolution strategy to map HSIs with arbitrary input spectral dimensions (C) to a fixed-dimensional feature space ($\widehat{C}$).

a spectral group size g and a stride k based on the input dimension C and our fixed number of groups s (where $s = 32$ in our implementation).

Specifically, we set the stride $k = \lfloor C/s \rfloor$ and the group size $g = \lceil C/s \rceil$. Since the group size g is always greater than or equal to the stride k (i.e., $\lceil C/s \rceil \geq \lfloor C/s \rfloor$), this strategy creates an overlapping sampling that ensures spectral continuity and robustness.

The ith independent convolutional layer $\mathcal{F}^i_{conv}$ (for $i = 0, \ldots, s-2$) processes the ith group of channels $\mathbf{X}_i$, which is a slice of g bands

$$\mathbf{X}_i = \mathbf{X}_{[\,i\cdot k\,:\,i\cdot k+g\,]}. \tag{1}$$

To ensure all spectral information is captured, the final group ($i = s-1$) processes all remaining channels from the last stride point

$$\mathbf{X}_{s-1} = \mathbf{X}_{[\,(s-1)\cdot k\,:\,C\,]}. \tag{2}$$

Each independent layer $\mathcal{F}^i_{conv}$ maps its input slice (which has a variable size g or $C-(s-1)k$) to a fixed-dimensional output $F_{out} = \widehat{C}/s$. The s output feature maps are then concatenated along the channel dimension to yield the final output $\mathbf{F}_{out} \in \mathbb{R}^{H\times W\times \widehat{C}}$. This "divide-and-conquer" approach decouples the model from specific sensor configurations. Each convolutional group learns to extract general-purpose information from its relative spectral range without needing to know the exact number of bands within it, thus enabling robust feature extraction when the available spectral channels are incomplete or inconsistent.

D. SHA Module

The SHA module [Fig. 4(c)] replaces static fusion with a dynamic, attention-based mechanism that learns which feature level to "trust" most at each spatial location. This is crucial because distinguishing subtle changes may require shallow-layer texture, while semantic changes require deep-layer context. Given the set of differential feature maps $\{\mathbf{D}_l\}_{l=1}^{L}$ as input, the SHA module first learns a spatial attention map $\mathbf{W}_l \in \mathbb{R}^{H_l\times W_l\times 1}$ for each level l

$$\mathbf{W}_l = \sigma_s\left(\mathcal{F}^{w,2}_{conv}\left(\sigma_r\left(\mathcal{F}^{w,1}_{conv}\left(\mathbf{D}_l\right)\right)\right)\right) \tag{3}$$

where σ_r is the ReLU activation and σ_s is the Sigmoid activation. This map highlights the importance of each spatial location at that specific level. Each feature map is then recalibrated via element-wise multiplication

$$\widetilde{\mathbf{D}}_l = \mathbf{D}_l \otimes \mathbf{W}_l \tag{4}$$

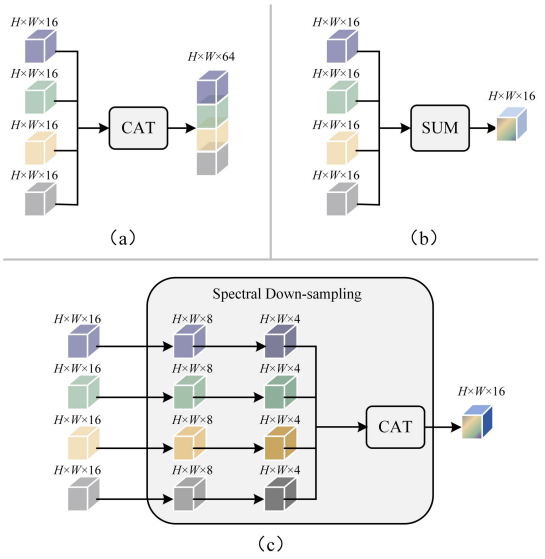


Fig. 4. Comparison of feature aggregation strategies. (a) Static fusion via concatenation. (b) Static fusion via summation. (c) Our proposed SHA, which dynamically fuses features using a spatial attention mechanism.

where $\otimes$ denotes element-wise multiplication with broadcasting, to emphasize salient regions. Finally, all recalibrated feature maps $\widetilde{\mathbf{D}}_l$ are upsampled to a uniform resolution, concatenated, and passed through a final fusion convolution $\mathcal{F}^{fuse}_{conv}$ to produce the aggregated feature map $\mathbf{Y}$

$$\mathbf{Y} = \mathcal{F}^{fuse}_{conv}\left(\left[\widetilde{\mathbf{D}}_1; \widetilde{\mathbf{D}}_2; \ldots; \widetilde{\mathbf{D}}_L\right]\right). \tag{5}$$

This process adaptively balances fine-grained details and semantic context, generating a "tailor-made" feature representation for each pixel that is critical for accurate multiclass discrimination.

E. Patch-to-Patch Semi-Supervised Training

To effectively train our deep architecture from limited labels, we employ a patch-to-patch semi-supervised strategy, as illustrated in Fig. 5 and summarized in Algorithm 1. This approach allows the network to learn from small, labeled image patches while leveraging vast unlabeled areas. Our strategy co-optimizes two complementary loss functions, based on the insight that while the Euclidean distance $\rho = \|\mathbf{X}_2 - \mathbf{X}_1\|_2$ is insufficient for multiclass discrimination, it serves as a cheap and effective heuristic for binary (change/no-change) discrimination.

First, we exploit this heuristic by computing a binary pseudo-loss ($\mathcal{L}_{binary}$) on all pixels within a training patch. We use K-Means clustering (K = 2) on ρ to generate binary pseudo-labels $\mathbf{Y}_{pseudo}$ and confidence scores $\mathbf{P}$. A confidence-weighted binary cross-entropy (BCE) loss is then computed

$$\mathcal{L}_{binary} = \frac{1}{\sum \mathbf{P}} \sum_{i,j} \mathbf{P}_{i,j} \cdot \mathrm{BCE}\left(\mathbf{Y}^{i,j}_{pseudo}, \hat{\mathbf{Y}}^{i,j}_{binary}\right). \tag{6}$$

This loss, applied to the entire patch, forces the encoder to learn a robust general-purpose feature representation from all available data.

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Algorithm 1 Patch-to-Patch Semi-Supervised Training

Input: HSI patch pair $\mathbf{x}_1, \mathbf{x}_2$; sparse multiclass labels $\mathbf{y}_{sup}$ and mask $\mathbf{M}$

Output: Trained D^{2}F^2 model θ

Initialize network parameters θ
// Pseudo-label Generation (on the patch)
Compute change magnitude $\rho = \|\mathbf{x}_2 - \mathbf{x}_1\|_2$ for **all** samples in the patch
Cluster ρ using K-means to obtain binary pseudo-labels $\mathbf{Y}_{pseudo}$ and confidence scores $\mathbf{P}$

// Alternate Optimization
repeat
  // Supervised Phase (on labeled pixels)
  Compute model output $\hat{\mathbf{Y}} = D^2F^2(\mathbf{x}_1, \mathbf{x}_2, \theta)$
  Compute multiclass loss: $\mathcal{L}_{multi} = \frac{1}{\sum \mathbf{M}} \sum_{i,j} \mathbf{M}_{i,j} \cdot \text{CE}(\mathbf{y}_{sup}^{i,j}, \hat{\mathbf{Y}}^{i,j})$
  Update parameters: $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{multi}$

  // Semi-supervised Phase (on all pixels)
  Compute model output $\hat{\mathbf{Y}} = D^2F^2(\mathbf{x}_1, \mathbf{x}_2, \theta)$
  Compute binary loss: $\mathcal{L}_{binary} = \frac{1}{\sum \mathbf{P}} \sum_{i,j} \mathbf{P}_{i,j} \cdot \text{BCE}(\mathbf{Y}_{pseudo}^{i,j}, \hat{\mathbf{Y}}_{binary}^{i,j})$
  Update parameters: $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{binary}$
until convergence

TABLE I
STRUCTURAL COMPARISON OF PCN AND FULLY CONVOLUTIONAL NETWORK (FCN) INFERENCE MODES

Model	PCN	FCN
Input layer	$W' \times H' \times C$	$W \times H \times C$
Hidden layer	$W' \times H' \times C/2$	$W \times H \times C/2$
Pooling layer	$W' \times H' \times N$	None
Output layer	$1 \times 1 \times N$	$W \times H \times N$

Concurrently, a standard multiclass supervised loss ($\mathcal{L}_{multi}$) is computed *only* on the sparse, manually labeled pixels within the patch (indicated by mask $\mathbf{M}$)

$$\mathcal{L}_{multi} = \frac{1}{\sum \mathbf{M}} \sum_{i,j} \mathbf{M}_{i,j} \cdot \text{CE}\left(\mathbf{Y}_{sup}^{i,j}, \hat{\mathbf{Y}}^{i,j}\right). \tag{7}$$

This loss uses the high-quality labels to fine-tune the network for fine-grained class separation. This alternating optimization creates a virtuous cycle where a strong binary extractor, trained on all data, is refined for the precise multiclass task using the sparse labels.

F. Map-to-Map Fully Convolutional Inference

A key advantage of the D^{2}F^2's fully convolutional (FCN) design is the decoupling of the training and inference paradigms, which enables significant computational gains. While training is efficiently performed patch-by-patch, the fully convolutional nature of the network allows inference to be performed in a single, efficient map-to-map pass.

This stands in sharp contrast to traditional patch-based convolutional network (PCN) (e.g., PCN, as shown in Table I, which must slice the input image into thousands of overlapping patches, run inference on each patch individually, and then stitch the results back together. This process incurs massive computational redundancy, as the features in the overlapping regions are recomputed many times.

Our FCN processes the entire $H \times W$ image in a single forward pass, producing the full change map at once. The ratio of computational cost (FLOPs) between a PCN and our FCN is approximately

$$\frac{\text{FLOPs}_{(PCN-total)}}{\text{FLOPs}_{(FCN)}} \approx \frac{H \cdot W}{H' \cdot W'} \tag{8}$$

where $H' \times W'$ is the PCN's input patch size. For a small 5×5 patch, our FCN is already ~25 times more computationally efficient. This gain is critical for processing large-scale satellite imagery in a practical timeframe.

IV. EXPERIMENTS

In this section, we first introduce the three public HSI datasets and evaluation criteria. We then conduct a series of experiments to comprehensively evaluate our proposed D^{2}F^2 framework. First, we compare D^{2}F^2 against a wide range of representative and SOTA methods to establish its superior performance. These include classic CVA-based methods (C^{2}VA [32], PCA-CVA [33], RCVA [34], and S^{2}CVA [51]); the classical machine learning method SVM; and a broad set of DL methods, including kPCAMNet [40], DSFA [52], DCVA [53], DNN, CNN, CSANet [21], MSDFFN [14], GETNET [13], CAST [54], GDAMamba [55], MSCSCNet [56], STMNet [57], and FDIEG-UNet [58]. Detailed ablation experiments are conducted to validate the effectiveness and design rationale of the core components in the proposed framework, including SAS, SHA, and the training patch size. The computational cost is further analyzed using both theoretical and practical indicators, including the number of parameters, checkpoint size, single-inference FLOPs, overall FLOPs, inference latency, throughput, peak GPU memory consumption, and mean epoch time.

A. Datasets and Image Preprocessing

The first two datasets are real-world HSI pairs from the Hyperion Imaging Spectrometer (EO-1 satellite) [9]. After preprocessing (spectrum, radiation, atmospheric, and geometric correction) to remove uncalibrated bands and noise, a total of 154 calibrated bands are used.

1) China Farmland: This 450 × 140 pixel dataset (Fig. 6) captures farmland changes near Yancheng, China, between 2006 and 2007. The ground truth contains an "unchanged" class and three distinct change classes (C1–C3).

2) USA Field: This 241 × 307 pixel dataset (Fig. 7) covers an irrigated agricultural area in Oregon, USA, with images from 2004 and 2007. This scene is more complex, with an "unchanged" class and six change classes (C1–C6).

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TABLE II
QUANTITATIVE EVALUATION OF EXPERIMENTAL RESULTS OF THE CHINA FARMLAND DATASET

Methods	OA (↑)	P	Pre	R	C0	C1	C2	C3	Kappa (↑)
C^2VA[32]	0.8678	0.5478	0.8731	0.7393	0.8905	0.6633	0.9045	-	0.7077
PCA-CVA[33]	0.8687	0.5460	0.8702	0.7463	0.8928	0.6604	0.9028	-	0.7107
RCVA[34]	0.8775	0.5598	0.9221	0.7176	0.8853	0.7281	0.9283	-	0.7217
S^2CVA[51]	0.8682	0.5472	0.8731	0.7393	0.8905	0.6957	0.9045	0.1643	0.7090
kPCAMNet[40]	0.8590	0.5459	0.8310	0.7509	0.8854	0.8603	0.7129	0.2718	0.6895
DSFA[52]	0.8708	0.4988	0.7780	0.8996	0.9508	0.7976	0.6252	-	0.7422
DCVA[53]	0.7398	**0.6101**	0.7105	0.3653	0.7634	0.6165	-	-	0.3326
SVM[62]	0.9204	0.5032	0.8065	0.9731	0.9792	0.7585	0.8655	0.2718	0.8397
CSANet[21]	0.8636	0.5317	0.8001	0.7680	0.8967	0.7811	0.7927	0.5475	0.7088
MSDFFN[14]	0.8716	0.5287	0.8105	0.7899	0.9051	0.8031	0.7954	0.4954	0.7275
Pixel-DNN	0.9052	0.4920	0.8259	0.9646	0.9857	0.6600	0.8658	0.2540	0.8133
Patch-CNN	0.9442	0.4994	0.8989	0.9803	**0.9963**	0.8422	0.8644	0.5285	0.8885
GETNET[13]	0.9671	0.5215	0.9403	0.9698	0.9827	0.8999	**0.9532**	**0.8217**	0.9312
CAST[54]	0.9114	0.5003	0.8364	0.9550	0.9781	0.6928	0.8608	0.6022	0.8227
GDAMamba[55]	0.9482	0.5197	0.9106	0.9473	0.9756	0.8644	0.9052	0.5000	0.8922
MSCSCNet[56]	0.8358	0.5405	0.8453	0.7443	0.8896	0.5169	0.8701	0.6905	0.6426
STMNet[57]	0.6146	0.4719	0.4382	0.9758	0.8171	0.5353	0.3419	-	0.2702
FDIEG-UNet[58]	0.8242	0.5409	0.8129	0.7694	0.8976	0.7076	0.6405	0.6667	0.6172
Ours(feature sum)	0.9655	0.5146	0.8931	**0.9818**	0.9918	0.8996	0.9225	0.6853	0.9289
Ours(feature cat)	0.9730	0.5193	0.9403	0.9524	0.9899	0.9206	0.9492	0.7400	0.9439
Ours(D^2F^2)	**0.9741**	0.5211	**0.9462**	0.9770	0.9885	**0.9483**	0.9443	0.7244	**0.9459**

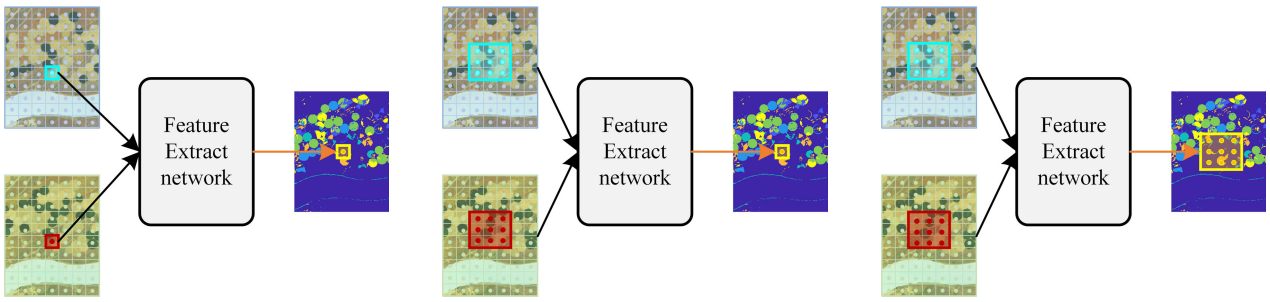


Fig. 5. Comparison of training paradigms for CD. (a) Pixel-to-pixel: maps one input pixel to one output label. (b) Patch-to-pixel: maps an input patch to a single label for the central pixel. (c) Patch-to-patch: maps an entire input patch to a corresponding output patch, as used in our FCN training.

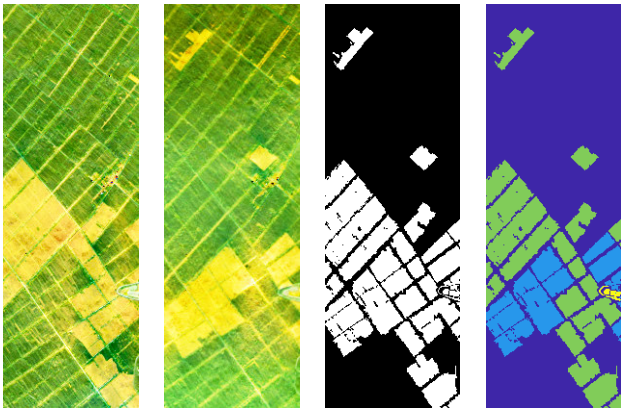


Fig. 6. Visualization results for the China Farmland dataset. (a) Image acquired in 2006. (b) Image acquired in 2007. (c) Binary ground truth. (d) Multiclass ground truth.

3) Simulated Dataset: This dataset (Fig. 8) was constructed from the Harvard HSI Database [59] (31 bands, 696 × 520 pixels) to test the model in a highly complex and controlled environment. We synthesized eight distinct change classes by swapping four different regions between the images. Critically, Gaussian noise (variance 0.1) was added to the post-change image to rigorously test the noise robustness of the compared algorithms.

B. Experimental Settings and Evaluation Criteria

All DL-based methods were implemented using the PyTorch framework [60] and trained on an NVIDIA GeForce 3090 GPU. We use the proposed semi-supervised framework, which leverages both labeled and unlabeled data as described in Section III. For the supervised component ($\mathcal{L}_{multi}$), we randomly select 20% of the available ground-truth pixels for training on the China Farmland and USA Field datasets, and 5% for the more densely labeled Simulated dataset. The remaining labeled pixels are used for validation. All pixels in the training patches are used to compute the binary pseudo-loss ($\mathcal{L}_{binary}$). The AdamW optimizer [61] is used with a learning rate of

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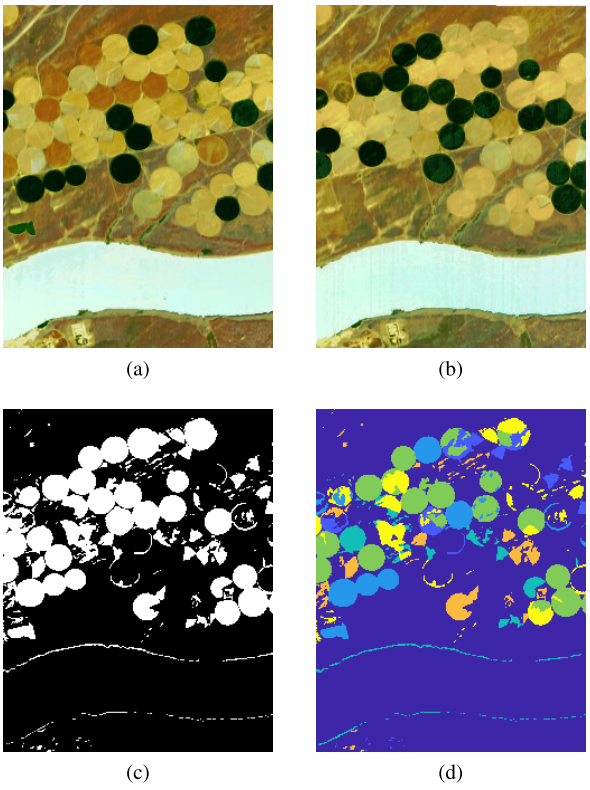


Fig. 7. Visualization results for the USA Field dataset. (a) Image acquired in 2004. (b) Image acquired in 2007. (c) Binary ground truth. (d) Multiclass ground truth.

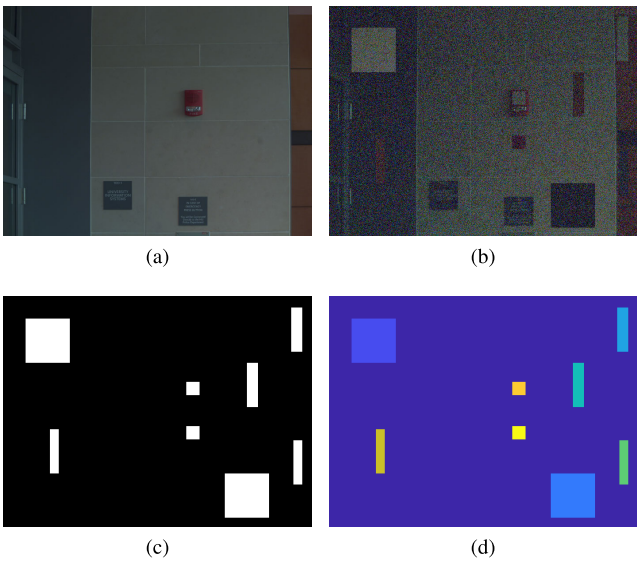


Fig. 8. Visualization results for the Simulated dataset. (a) Pre-changed images. (b) Post-changed images. (c) Binary ground truth. (d) Multiclass ground truth.

0.0003, weight decay of 0.01, and a batch size of 64 for 10 epochs.

We evaluate performance both qualitatively (visual change maps) and quantitatively. For quantitative evaluation, we use overall accuracy (OA), precision (Pre), recall (R), and the Kappa Coefficient (*Kappa*), which are standard metrics for assessing the consistency between the detection map and the ground truth. Specifically, the OA is defined as

$$OA = \frac{TP + TN}{TP + TN + FP + FN} \tag{9}$$

and the variables TP, TN, FP, and FN represent the true positives, the true negatives, the false positives, and the false negatives, respectively. Precision measures the reliability of detected changed samples, while Recall measures the ability to identify actual changed samples. They are defined as

$$Precision = \frac{TP}{TP + FP} \tag{10}$$

$$Recall = \frac{TP}{TP + FN}. \tag{11}$$

In the CD task, the kappa coefficient indicates the consistency between the CD map and the ground-truth map. The larger the value of the kappa coefficient, the better the performance of the corresponding method. The kappa coefficient is denoted as

$$Kappa = \frac{OA - P}{1 - P} \tag{12}$$

where

$$P = \frac{(TP + FP)(TP + FN) + (TN + FN)(FP + TN)}{(TP + TN + FP + FN)^2}. \tag{13}$$

C. Experiments on the China Farmland Dataset

The quantitative results on the China Farmland dataset (Table II) and qualitative maps (Fig. 9) demonstrate the clear superiority of our method. The unsupervised CVA-based methods (a)–(d) and other unsupervised DL methods (e)–(g) largely fail, struggling to identify all three change classes and producing noisy results, such as DCVA and kPCAMNet. This confirms our problem statement that robust unsupervised multiclass CD is extremely challenging. The supervised methods show marked improvement, but still exhibit key weaknesses. The pixel-based Pixel-DNN (k) suffers from intense salt-and-pepper noise, highlighting the necessity of spatial context. The patch-based Patch-CNN (l), while cleaner, produces characteristic blocky artifacts and ambiguous boundaries, a known flaw of patch-to-pixel inference. Among the stronger DL baselines, GETNET (m) produces smoother regions but still fails to accurately delineate several field boundaries. The SOTA models, namely CAST (n), GDAMamba (o), MSCSCNet (p), STMNet (q), and FDIEG-UNet (r), further enrich the comparison with recent advanced architectures. Their visual results indicate that recent network designs can improve regional consistency to some extent, but they remain sensitive to boundary ambiguity, small-area changes, and class confusion. In particular, MSCSCNet and STMNet show noticeable fragmented or distorted predictions, while CAST, GDAMamba, and FDIEG-UNet tend to miss detailed parcel structures.

In contrast, our D^2F^2 outperforms in terms of OA and Kappa, achieving the highest OA (0.9741) and Kappa (0.9459). We attribute this superior performance to our core design choices. First, unlike the more static fusion in GETNET, our SHA module dynamically weights hierarchical features, allowing it to better distinguish spectrally similar classes like C1 and C2 (achieving high per-class accuracies of 0.9483 and 0.9443). Second, as seen in Fig. 9(s), our map-to-map inference produces a spatially coherent result with the sharpest boundaries and least noise, demonstrating the practical benefit of our fully convolutional design.

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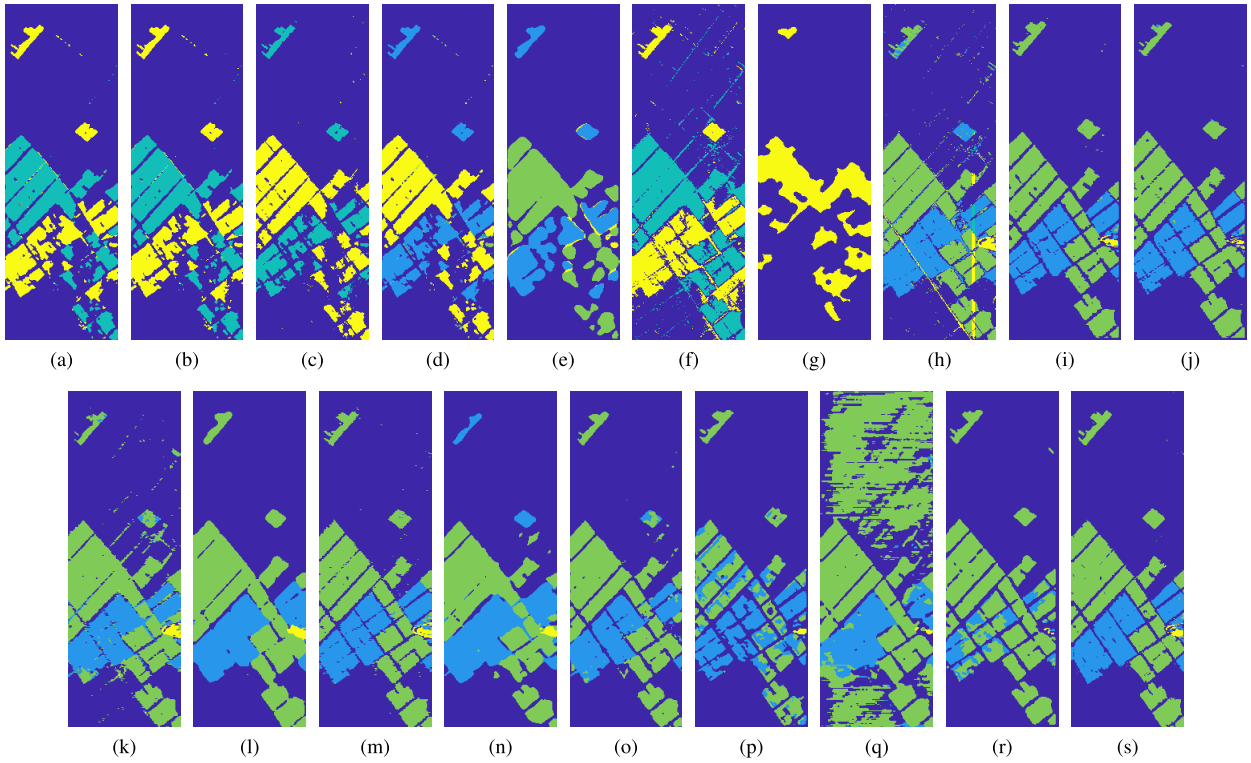


Fig. 9. CD results for the China Farmland dataset. (a) C^2VA. (b) PCA-CVA. (c) RCVA. (d) S^2CVA. (e) kPCAMNet. (f) DSFA. (g) DCVA. (h) SVM. (i) CSANet. (j) MSDFFN. (k) DNN. (l) CNN. (m) GETNET. (n) CAST. (o) GDAMamba. (p) MSCSCNet. (q) STMNet. (r) FDIEG-UNet. (s) Ours.

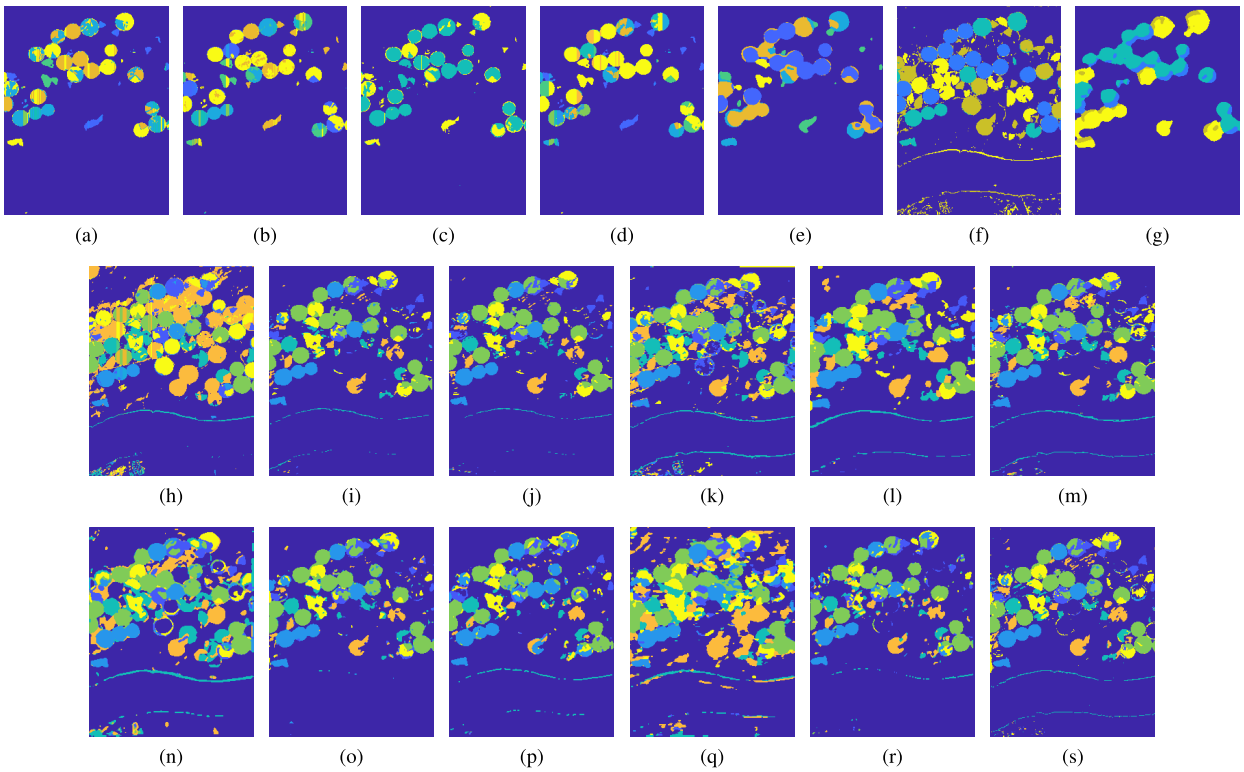


Fig. 10. CD results for the USA Field dataset. (a) C^2VA. (b) PCA-CVA. (c) RCVA. (d) S^2CVA. (e) kPCAMNet. (f) DSFA. (g) DCVA. (h) SVM. (i) CSANet. (j) MSDFFN. (k) DNN. (l) CNN. (m) GETNET. (n) CAST. (o) GDAMamba. (p) MSCSCNet. (q) STMNet. (r) FDIEG-UNet. (s) Ours.

D. Experiments on the USA Field Dataset

Table III presents the quantitative results for the USA Field dataset. This scene represents a more challenging scenario, with multiple change classes and evident sensor striping noise. Fig. 10 further visualizes the qualitative results and shows the difficulty of accurately identifying different crop-change categories under noisy observation conditions. The unsupervised

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TABLE III
QUANTITATIVE EVALUATION OF EXPERIMENTAL RESULTS OF THE USA FIELD DATASET

Methods	OA (↑)	P	Pre	R	C0	C1	C2	C3	C4	C5	C6	Kappa (↑)
C^2VA[32]	0.8330	0.6664	0.9812	0.6501	0.9066	0.5277	0.2894	0.0511	0.7458	-	-	0.4995
PCA-CVA[33]	0.8409	0.6659	0.9884	0.6499	0.9067	0.2243	0.4898	0.018	0.9421	-	-	0.5238
RCVA[34]	0.8050	**0.6726**	**0.9959**	0.6084	0.8969	0.1010	0.2936	0.0887	0.2895	0.4928	-	0.4045
S^2CVA[51]	0.8118	0.6646	0.9812	0.6501	0.9066	0.2666	0.1978	0.0510	0.5036	0.4023	0.3331	0.4391
kPCAMNet[40]	0.8453	0.6500	0.9153	0.6930	0.9166	0.4603	0.7430	0.2567	0.8416	0.2352	-	0.5580
DSFA[52]	0.7865	0.5496	0.7269	0.9075	0.2963	0.2527	0.1869	**0.9688**	0.7371	-	-	0.3326
DCVA[53]	0.8111	0.6375	0.8151	0.6793	0.9106	0.1159	0.6601	0.2100	0.1845	-	-	0.4790
SVM[62]	0.8502	0.5462	0.5322	0.9542	0.9814	0.5190	0.8215	0.3750	0.9256	0.3011	0.3543	0.6699
CSANet[21]	0.8496	0.6209	0.7378	0.6889	0.9109	0.4478	0.6820	0.4391	0.7432	0.5174	0.5580	0.6033
MSDFFN[14]	0.8505	0.5287	0.7425	0.6850	0.9100	0.4111	0.7113	0.4563	0.7227	0.5022	0.6022	0.6036
Pixel-DNN	0.8660	0.5403	0.7147	0.9634	0.9913	0.5811	0.7903	0.3334	0.8923	0.3180	0.6891	0.7084
Patch-CNN	0.8922	0.5454	0.7145	0.9709	0.9915	0.6670	0.7338	0.4957	0.8536	0.4725	0.6633	0.7628
GETNET[13]	0.9155	0.5662	0.8032	**0.9753**	**0.9930**	0.7552	0.7381	0.7506	0.9342	0.5883	0.5192	0.8051
CAST[54]	0.8238	0.5116	0.6116	0.9685	0.9889	0.4798	0.6906	0.2736	0.8022	0.2764	0.7110	0.6392
GDAMamba[55]	0.9323	0.6207	0.9307	0.8712	0.9631	0.6056	0.8360	0.6996	0.8999	**0.7815**	**0.8248**	0.8216
MSCSCNet[56]	0.8584	0.6170	0.7980	0.7612	0.9312	0.4303	0.4750	0.4541	0.8232	0.6016	0.5731	0.6304
STMNet[57]	0.7990	0.5721	0.5412	0.9531	0.9400	0.3766	0.6027	0.1463	0.6592	0.3008	0.3776	0.5302
FDIEG-UNet[58]	0.8388	0.6272	0.7446	0.6680	0.9059	0.3331	0.6272	0.3967	0.6948	0.4332	0.5789	0.5677
Ours(feature sum)	0.9482	0.5950	0.8652	0.9402	0.9864	0.7628	0.9140	0.7522	0.9120	0.7399	0.7544	0.8721
Ours(feature cat)	0.9523	0.5987	0.8841	0.9134	0.9857	**0.8505**	**0.9351**	0.7061	**0.9611**	0.7407	0.7510	0.8811
Ours(D^2F^2)	**0.9528**	0.5983	0.8931	0.9563	0.9863	0.7683	0.9235	0.7234	0.9540	0.7407	0.7838	**0.8824**

methods (a)–(g) show clear limitations in this complex scene, as they fail to capture the full range of change classes and often produce incomplete or noisy maps. The traditional SVM method (h) is also sensitive to the striping noise, which leads to obvious artifacts in the detection result. Among the earlier DL baselines, CSANet (i), MSDFFN (j), Pixel-DNN (k), Patch-CNN (l), and GETNET (m) improve the overall regional consistency, but salt-and-pepper noise, boundary ambiguity, and class confusion are still visible in several local areas.

The SOTA models, including CAST (n), GDAMamba (o), MSCSCNet (p), STMNet (q), and FDIEG-UNet (r), provide stronger comparison with recent advanced architectures. Their results show that recent models can better preserve several major changed regions than traditional and early DL baselines. However, they still suffer from different limitations in this noisy scene. CAST and GDAMamba show local omissions and mixed class assignments around dense crop regions, while MSCSCNet and STMNet produce fragmented predictions or striping-related artifacts. FDIEG-UNet obtains relatively compact regions, but some small changed areas and boundaries remain inaccurate.

In contrast, the proposed D^2F^2 (s) achieves the best overall quantitative performance, with the highest OA of 0.9528 and Kappa of 0.8824. It also maintains robust per-class identification across the six change categories and effectively suppresses sensor noise while preserving clearer boundaries. These results indicate that the proposed map-to-map inference and dynamic feature fusion strategy can provide more stable multiclass CD performance in complex and noisy agricultural scenes.

E. Experiments on the Simulated Dataset

The quantitative and qualitative results for the Simulated dataset are presented in Table IV and Fig. 11, respectively. This dataset was designed to evaluate model robustness under high interclass similarity and strong additive noise. The unsupervised methods (a)–(g) face considerable challenges in this setting, with most of them failing to identify multiple change classes. In particular, DSFA (f) is severely affected by noise and fails to produce a meaningful change map. The supervised SVM method (h) is also sensitive to noise interference, leading to obvious misclassified regions in the detection result. Among the earlier DL baselines, CSANet (i), MSDFFN (j), Pixel-DNN (k), Patch-CNN (l), and GETNET (m) achieve improved structural consistency, but they still exhibit visible noise sensitivity, local false alarms, and incomplete boundary recovery.

The SOTA models, including CAST (n), GDAMamba (o), MSCSCNet (p), STMNet (q), and FDIEG-UNet (r), provide stronger comparison under this challenging simulated scenario. These methods generally produce cleaner maps than conventional and early DL baselines, indicating their improved representation ability. However, different degrees of noise-induced errors and class confusion can still be observed. For example, STMNet (q) shows evident noisy responses and fragmented predictions, while FDIEG-UNet (r) still introduces local false detections around several changed regions. These results suggest that strong additive noise remains challenging even for recent advanced architectures.

In contrast, the proposed D^2F^2 framework (s) achieves the best overall performance, with an OA of 0.9982 and a Kappa of 0.9888. As shown in Fig. 11(s), the proposed method produces a clean and spatially coherent change map with sharper boundaries and fewer isolated noisy predictions. This indicates that the proposed feature extraction and dynamic fusion strategy can effectively capture the underlying change signatures while suppressing strong random noise.

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TABLE IV
QUANTITATIVE EVALUATION OF EXPERIMENTAL RESULTS OF THE SIMULATED DATASET

Methods	OA (↑)	P	Pre	R	C0	C1	C2	C3	C4	C5	C6	C7	C8	Kappa (↑)
C^2VA[32]	0.9597	0.8596	**0.9999**	0.7284	0.9752	0.9712	**1.0000**	0.5089	-	-	-	-	-	0.7128
PCA-CVA[33]	0.9577	0.8580	**0.9999**	0.7448	0.9770	0.9225	0.9723	0.6176	-	-	-	-	-	0.7020
RCVA[34]	0.9302	0.8846	0.9824	0.3912	0.9465	0.8758	-	0.4672	-	-	-	-	-	0.3951
S^2CVA[51]	0.9734	0.8599	**0.9999**	0.7284	0.9752	0.9416	0.9999	0.6827	-	-	-	-	-	0.8100
kPCAMNet[40]	0.9666	0.8579	0.9827	0.7350	0.9762	0.7770	0.9649	0.1372	-	-	-	-	-	0.7646
DSFA[52]	0.5835	0.4996	0.1111	0.9790	0.0003	0.9807	0.5315	0.1735	-	-	-	-	-	0.1678
DCVA[53]	0.9295	0.9019	0.9990	0.1714	0.9284	0.9990	-	-	-	-	-	-	-	0.2808
SVM[62]	0.6685	0.5397	0.2164	0.9981	**0.9999**	0.9993	0.9928	0.5259	0.2441	0.1131	0.0207	0.6055	0.3652	0.2799
CSANet[21]	0.9844	0.8383	0.9069	0.9122	0.9918	0.9411	0.9424	0.8595	0.8105	0.8682	0.8600	0.8145	0.7919	0.9038
MSDFFN[14]	0.9841	0.8392	0.9100	0.9043	0.9911	0.9420	0.9408	0.8528	0.8158	0.8925	0.8364	0.8073	0.9028	0.9013
Pixel-DNN	0.9813	0.8308	0.8625	0.9511	0.9954	0.9929	0.9999	0.8569	0.8542	0.3502	0.4320	0.9803	0.7628	0.8895
Patch-CNN	0.9949	0.8348	0.9470	0.9958	0.9996	0.9775	0.9409	0.9256	0.9215	0.9701	0.9333	0.9325	0.8330	0.9692
GETNET[13]	0.9847	0.8380	0.9144	0.9232	0.9929	0.9913	0.9923	0.8037	0.8553	0.7676	0.3847	0.8955	0.9375	0.9058
CAST[54]	0.9973	0.8367	0.9728	**0.9987**	**0.9999**	0.9869	**1.0000**	0.9276	0.9339	0.8846	0.9642	0.8870	**1.0000**	0.9832
GDAMamba[55]	0.9976	0.8392	0.9967	0.9909	0.9992	0.9987	0.9946	**0.9932**	0.8528	0.9801	**0.9932**	0.9900	0.9128	0.9849
MSCSCNet[56]	0.9879	0.8403	0.9485	0.9291	0.9934	0.9535	0.9697	0.8882	0.8539	0.9395	0.8879	0.6390	**1.0000**	0.9241
STMNet[57]	0.9149	**0.9149**	0.5934	0.9711	0.9149	-	-	-	-	-	-	-	-	-
FDIEG-UNet[58]	0.9517	0.8715	0.9403	0.5344	0.9584	-	0.9718	0.8649	**1.0000**	0.4642	-	0.8398	0.5761	0.6245
Ours(D^2F^2)	**0.9982**	0.8388	0.9931	0.9919	0.9993	**0.9998**	**1.0000**	0.9416	0.9543	**0.9822**	0.9744	**0.9955**	0.9529	**0.9888**

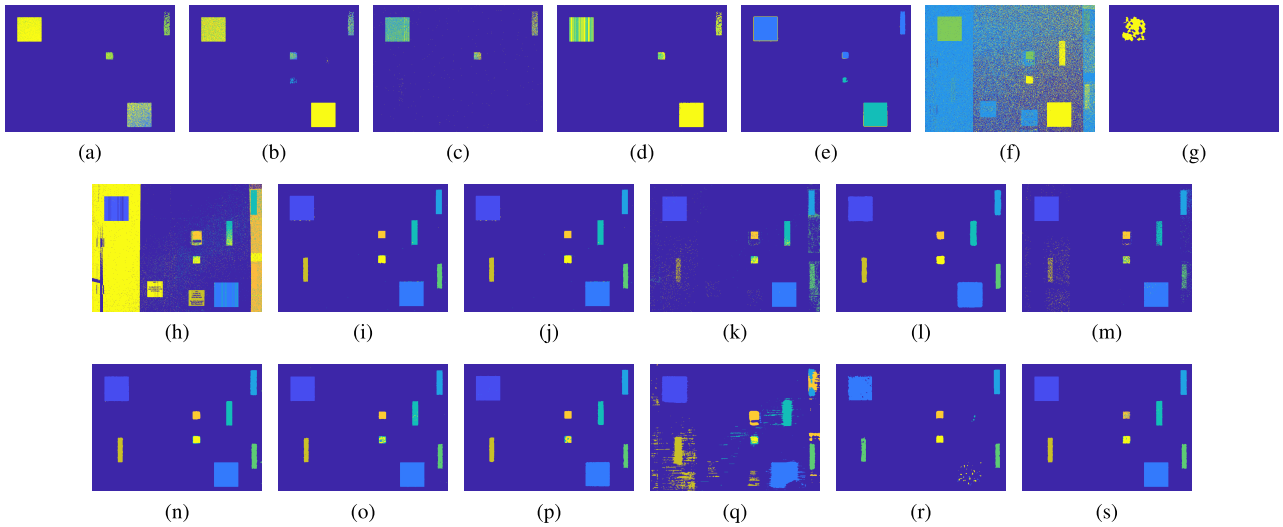


Fig. 11. CD results for the simulated dataset. (a) C^2VA. (b) PCA-CVA. (c) RCVA. (d) S^2CVA. (e) kPCAMNet. (f) DSFA. (g) DCVA. (h) SVM. (i) CSANet. (j) MSDFFN. (k) DNN. (l) CNN. (m) GETNET. (n) CAST. (o) GDAMamba. (p) MSCSCNet. (q) STMNet. (r) FDIEG-UNet. (s) Ours.

F. Ablation Studies and Parameter Analysis

1) Ablation on Spectral Inconsistency (SAS Module): Table V provides direct and compelling evidence for the robustness of our SAS module. This experiment, designed to simulate the spectral inconsistency challenge, takes the fully trained model and randomly drops spectral bands during inference (the model was never trained on this "damaged" data).

The results are remarkable. Even with 34 spectral bands (over 20% of the total spectral information) removed, the performance drop is negligible (e.g., Kappa on the CN dataset drops by only 0.003, from 0.94259 to 0.93882). This stability demonstrates that the SAS module's grouped sampling strategy successfully learns a stable, sensor-agnostic feature representation from relative spectral ranges, rather than merely "memorizing" fixed-band locations. This confirms its effectiveness in handling missing-channel and band-mismatch settings.

TABLE V
CD RESULTS WITH MISSING SPECTRAL CHANNELS

Channels	CN		US	
	OA (↑)	Kappa (↑)	OA (↑)	Kappa (↑)
120	0.97056	0.93882	0.95351	0.88457
130	0.97129	0.94041	0.95502	0.88814
140	0.96906	0.93592	0.95344	0.88467
150	0.97175	0.94125	0.95441	0.88682
154	0.97241	0.94259	0.95433	0.88667

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TABLE VI
EFFECT OF DIFFERENT TRAINING PATCH SIZES ON KAPPA VALUES

Patch Size	7	11	15	19	23
CN Dataset	0.74728	0.94122	**0.94592**	0.93452	0.93801
US Dataset	0.55618	0.85174	**0.88241**	0.87089	0.87854

2) Ablation on Feature Fusion (SHA Module): This experiment validates our hypothesis that dynamic feature fusion (SHA) outperforms static fusion (SUM, CAT). The results are shown in the last three rows of Tables II and III. "Ours (feature sum)" performs the worst, confirming our hypothesis that element-wise summation leads to destructive feature conflation. "Ours (feature cat)," while a strong baseline, is still consistently outperformed by our full D^2F^2 model (using SHA) (e.g., 0.9439 versus 0.9459 Kappa on CN). More importantly, this superior accuracy is also achieved with greater parameter efficiency, as further analyzed in Section IV-G.

3) Analysis of Training Patch Size: Table VI explores the impact of the training patch size. A clear trend emerges: performance is poor at 7 × 7 (Kappa 0.747 on CN), increases dramatically at 11 × 11, and saturates around 15 × 15 (0.9459 Kappa). This confirms two points: 1) The model requires a sufficient receptive field (at least 11 × 11) to leverage spatial context, which is the key weakness of pixel-based methods and 2) Performance does not significantly increase (and even slightly drops) at 19 × 19 or 23 × 23. This suggests a tradeoff. While larger patches see more context, they may also average the loss signal over overly heterogeneous regions, hindering fine-grained boundary learning. Therefore, we select 15 × 15 as the optimal balance between spatial context and training efficiency.

4) Ablation on the Number of Channels: Table VII first investigates the influence of the channel number. The channel dimension controls the representation capacity of the feature extractor. When the number of channels is set to 96, the model achieves an OA of 0.9715 and a Kappa of 0.9403, indicating that a relatively compact feature space can already capture useful change information. Increasing the channel number to 128 further improves the performance, reaching the best OA of 0.9741 and Kappa of 0.9459.

Further increasing the channel number to 192 or 256 does not bring additional gains. Instead, the Kappa values decrease to 0.9301 and 0.9378. This suggests that simply enlarging the feature dimension may introduce redundant or less stable responses, which can weaken the discrimination of subtle multiclass changes. Therefore, 128 channels are selected as the default setting, providing the best balance between representation capacity and model stability.

5) Ablation on the Number of Groups: The second part of Table VII evaluates the effect of the number of groups. The grouping strategy is introduced to encourage structured spectral–spatial representation and reduce excessive feature coupling. When the number of groups is set to 1, 8, and 16, the corresponding Kappa values are 0.9282, 0.9417, and 0.9385, respectively. These results indicate that grouped modeling can improve the representation to some extent, but insufficient or suboptimal grouping may limit the ability to model diverse spectral responses.

The best performance is achieved when the number of groups is set to 32, with an OA of 0.9741 and a Kappa of 0.9459. Compared with the settings using fewer groups, this configuration provides a more stable and discriminative feature representation. This result supports the use of grouped feature modeling in the proposed framework.

6) Ablation on the Group Branch: The third part of Table VII directly examines the contribution of the group branch. When the group branch is removed, the model achieves an OA of 0.9712 and a Kappa of 0.9400. Although this result remains competitive, it is still lower than the full setting with the group branch, which achieves an OA of 0.9741 and a Kappa of 0.9459.

This comparison confirms that the group branch contributes to multiclass CD. By explicitly organizing features into groups, the model can better exploit complementary spectral–spatial information and reduce feature interference across different change categories. The performance gap between the two settings indicates that the group branch is not redundant, but helps improve the stability and discriminative ability of the learned representation.

7) Ablation on the Pseudo-Label Ratio: The fourth part of Table VII studies the influence of the pseudo-label ratio. Without pseudo-label supervision, the model only obtains an OA of 0.8999 and a Kappa of 0.7873, indicating that the available labeled samples alone are insufficient to fully guide the model in the multiclass CD task. When the pseudo-label ratio is increased to 0.01 and 0.05, the performance improves notably, reaching Kappa values of 0.9079 and 0.9439, respectively. This shows that pseudo-labels can provide useful auxiliary supervision and help the model exploit unlabeled samples more effectively.

The best result is obtained when the pseudo-label ratio is set to 0.2, where the OA and Kappa reach 0.9741 and 0.9459, respectively. This indicates that a sufficient amount of reliable pseudo-label information can enhance model training. However, the performance at 0.1 is lower than that at 0.05 and 0.2, suggesting that pseudo-label supervision is sensitive to the balance between useful additional samples and potential label noise. Based on the overall comparison, 0.2 is adopted as the preferred setting in this experiment.

8) Ablation on Pseudo-Label Perturbation: The last part of Table VII analyzes the pseudo-label perturbation strategy, where each setting is reported as Flip/Shift_h/Shift_w. The baseline setting without perturbation, namely 0.00/0/0, achieves the best performance, with an OA of 0.9741 and a Kappa of 0.9459. This indicates that the generated pseudo-labels are already sufficiently stable under the current training configuration, and additional perturbation is not necessary for this dataset.

When flip perturbation is introduced, the performance generally decreases. For example, the Kappa values drop to 0.9270, 0.9279, 0.9440, and 0.9239 when the flip ratio is set to 0.01, 0.03, 0.05, and 0.10, respectively. Spatial shifts also lead to performance degradation, with Kappa values ranging from 0.9144 to 0.9348. These results suggest that

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TABLE VII
ABLATION STUDIES ON THE CHINA FARMLAND DATASET

Ablation Factor	Metric	V1	V2	V3	V4	V5	V6	V7	V8	V9
Channel Number	Setting	96	128	192	256	–	–	–	–	–
	OA	0.9715	**0.9741**	0.9660	0.9699	–	–	–	–	–
	Kappa	0.9403	**0.9459**	0.9301	0.9378	–	–	–	–	–
Number of Groups	Setting	1	8	16	32	–	–	–	–	–
	OA	0.9652	0.9721	0.9704	**0.9741**	–	–	–	–	–
	Kappa	0.9282	0.9417	0.9385	**0.9459**	–	–	–	–	–
Group Branch	Setting	✓	×	–	–	–	–	–	–	–
	OA	**0.9741**	0.9712	–	–	–	–	–	–	–
	Kappa	**0.9459**	0.9400	–	–	–	–	–	–	–
Pseudo Ratio	Setting	0.0	0.01	0.05	0.1	0.2	–	–	–	–
	OA	0.8999	0.9552	0.9731	0.9513	**0.9741**	–	–	–	–
	Kappa	0.7873	0.9079	0.9439	0.8990	**0.9459**	–	–	–	–
Pseudo-Label Perturbation	Setting	0.00/0/0	0.01/0/0	0.03/0/0	0.05/0/0	0.10/0/0	0.00/1/0	0.00/2/0	0.00/0/1	0.00/0/2
	OA	**0.9741**	0.9647	0.9653	0.9731	0.9636	0.9685	0.9582	0.9680	0.9687
	Kappa	**0.9459**	0.9270	0.9279	0.9440	0.9239	0.9345	0.9144	0.9336	0.9348

* For pseudo-label perturbation, each setting is reported as Flip/Shift$_h$/Shift$_w$.

TABLE VIII
COMPARISON OF PARAMETERS AND FLOPS OF DIFFERENT ALGORITHMS

Methods	Input Shape	Repeat	Params (M)	Ckpt. (MB)	Single FLOPs (G)	Overall FLOPs	Latency (ms/img)	Throughput (img/s)	Peak Mem. (GB)	Epoch Time (s)
CSANet[21]	$2 \times 154 \times 9 \times 9$	16384	40.420	154.227	0.280	4.588T	0.259	3861.764	0.635	2.438
MSDFFN[14]	$2 \times 154 \times 9 \times 9$	16384	26.459	101.070	2.504	41.026T	0.690	1450.300	2.422	4.622
Pixel-DNN	$2 \times 154 \times 1 \times 1$	16384	0.164	0.630	**0.0003**	4.915G	**0.019**	**52890.100**	0.065	2.949
Patch-CNN	$2 \times 154 \times 15 \times 15$	16384	0.368	1.412	0.328	5.374T	0.061	16362.750	0.208	3.411
GETNET[13]	$2 \times 154 \times 1 \times 1$	16384	92.182	351.661	1.045	17.121T	1.518	658.677	1.952	5.430
CAST[54]	$2 \times 154 \times 7 \times 7$	16384	0.312	1.213	0.026	0.426T	0.133	7504.097	0.085	**1.618**
GDAMamba[55]	$2 \times 154 \times 5 \times 5$	16384	**0.103**	**0.428**	0.003	49.152G	0.438	2285.637	**0.049**	4.082
MSCSCNet[56]	$2 \times 154 \times 11 \times 11$	16384	9.796	37.430	29.793	488.129T	1.831	546.067	0.940	9.481
STMNet[57]	$2 \times 154 \times 31 \times 31$	16384	44.906	171.437	10.416	170.656T	0.962	1039.231	4.823	8.988
FDIEG-UNet[58]	$2 \times 154 \times 9 \times 9$	16384	1.404	5.402	0.197	3.228T	0.392	2553.310	0.339	3.319
Ours(feature sum)	$2 \times 154 \times 128 \times 128$	1	0.199	0.779	0.176	**0.176G**	0.117	8514.796	0.145	54.385
Ours(feature cat)	$2 \times 154 \times 128 \times 128$	1	0.243	0.948	0.195	0.195G	0.126	7916.089	0.146	21.800
Ours(D^2F^2)	$2 \times 154 \times 128 \times 128$	1	0.203	0.800	0.177	0.177G	0.141	7098.370	0.145	5.772

excessive perturbation may disturb the spatial correspondence between pseudo-labels and image structures, especially near object boundaries. Therefore, the nonperturbed pseudo-label setting is retained, as it provides the most reliable supervision and preserves the spatial consistency required for accurate multiclass CD.

G. Network Parameters and FLOPs

Finally, we analyze the computational complexity, which highlights the practical advantage of our FCN design. Table VIII compares our map-to-map model with patch-based competitors. More importantly, this superior accuracy is achieved with greater parameter efficiency. As shown in Table VIII, the SHA model (0.203M params) is more compact than the "Feature Cat" model (0.243M params). This confirms that our SHA module is not just more accurate, but also more efficient, by replacing brute-force concatenation with an intelligent, attention-based selection mechanism.

In addition to parameters and FLOPs, Table VIII further reports checkpoint size, latency, throughput, peak GPU memory, and mean epoch time. For a fair comparison, patch-based methods are evaluated with 16 384 repeated patch-wise inferences, whereas our FCN-based variants require only one forward pass for whole-image prediction.

The Overall FLOPs column clearly shows the efficiency gap. CSANet, MSDFFN, GETNET, and STMNet require 4.588T, 41.026T, 17.121T, and 170.656T FLOPs, respectively. In contrast, Ours (D^2F^2) requires only 0.177 GFLOPs, corresponding to reductions of approximately 2.59×10^4, 2.32×10^5, 9.67×10^4, and 9.64×10^5, respectively. Meanwhile, the proposed model contains only 0.203M parameters, has a compact 0.800-MB checkpoint, and uses only 0.145-GB peak GPU memory.

Although Pixel-DNN achieves the lowest latency and highest throughput, its pixel-wise design lacks spatial contextual modeling and leads to weaker detection performance. Overall, the proposed D^2F^2 provides a more favorable tradeoff among accuracy, total computational cost, memory usage, and practical deployment efficiency.

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V. Conclusion

In this article, we proposed the D^2F^2 framework, a holistic and end-to-end solution designed to systematically address three of the most critical and persistent challenges in multiclass HSI-CD: spectral inconsistency, inefficient feature fusion, and label scarcity.

Our framework's contributions are threefold. First, the SAS module improves robustness to spectral inconsistency caused by missing channels and varying band availability. By adaptively remapping inputs of arbitrary band counts to a unified feature space, it enables, for the first time, a single end-to-end model to be practically deployed across diverse and inconsistent HSI datasets. Second, the SHA module replaces the static and suboptimal fusion methods common in the literature. Through its dynamic attention mechanism, the SHA module intelligently aggregates differential features from multiple semantic levels, creating highly discriminative representations essential for separating spectrally similar change classes. Third, our novel semi-supervised training strategy addresses the critical bottleneck of label scarcity. By coupling a patch-to-patch training scheme with a robust binary pseudo-loss, our model learns effectively from limited annotations. Crucially, this fully convolutional design transitions to an efficient map-to-map inference mode, which eliminates the massive computational redundancy of patch-based methods and achieves a remarkable speedup, moving the framework from a theoretical model to a practical, deployable system. Comprehensive experiments on two datasets validate our approach. The D^2F^2 framework is proven to not only outperform current SOTA methods in detection accuracy (both OA and Kappa) but also to be orders of magnitude faster in inference speed.

Future work will focus on extending the proposed framework to more complex change scenarios, improving its robustness under more limited supervision and domain shift, and exploring the integration of foundation-model-based hyperspectral representations to further enhance its generalization ability and practical applicability.

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